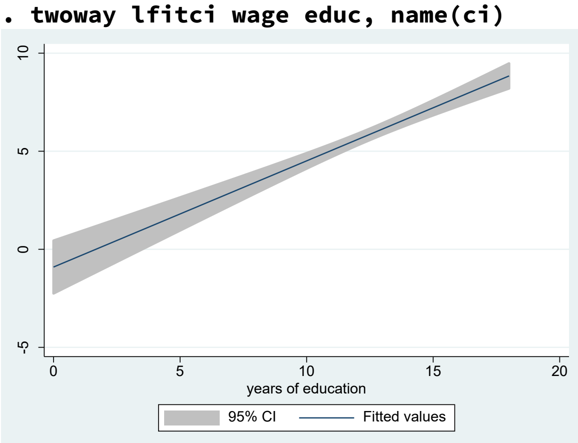
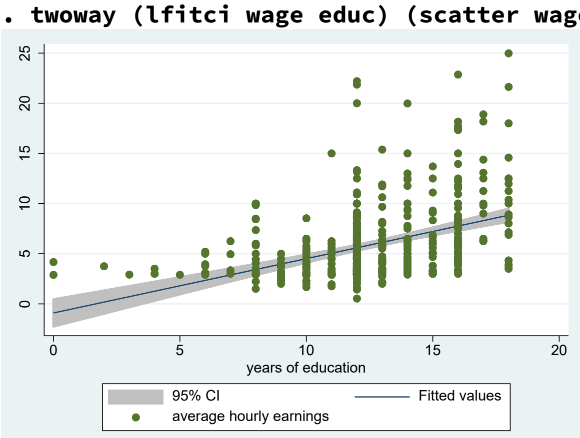
```
. /* Example 2.4
>    Data from 1976
>     . constant slope
>     . forecast near edu=0
> */
.
. clear all
.
. use wage1, clear
. desc

Contains data from wage1.dta
 Observations:           526
    Variables:            24                  16 Sep 1996 15:52
--------------------------------------------------------------------------------
Variable      Storage   Display    Value
    name         type    format    label      Variable label
--------------------------------------------------------------------------------
wage            float   %8.2g                 average hourly earnings
educ            byte    %8.0g                 years of education
exper           byte    %8.0g                 years potential experience
tenure          byte    %8.0g                 years with current employer
nonwhite        byte    %8.0g                 =1 if nonwhite
female          byte    %8.0g                 =1 if female
married         byte    %8.0g                 =1 if married
numdep          byte    %8.0g                 number of dependents
smsa            byte    %8.0g                 =1 if live in SMSA
northcen        byte    %8.0g                 =1 if live in north central U.S
south           byte    %8.0g                 =1 if live in southern region
west            byte    %8.0g                 =1 if live in western region
construc        byte    %8.0g                 =1 if work in construc. indus.
ndurman         byte    %8.0g                 =1 if in nondur. manuf. indus.
trcommpu        byte    %8.0g                 =1 if in trans, commun, pub ut
trade           byte    %8.0g                 =1 if in wholesale or retail
services        byte    %8.0g                 =1 if in services indus.
profserv        byte    %8.0g                 =1 if in prof. serv. indus.
profocc         byte    %8.0g                 =1 if in profess. occupation
clerocc         byte    %8.0g                 =1 if in clerical occupation
servocc         byte    %8.0g                 =1 if in service occupation
lwage           float   %9.0g                 log(wage)
expersq         int     %9.0g                 exper^2
tenursq         int     %9.0g                 tenure^2
--------------------------------------------------------------------------------
Sorted by:

. sum wage lwage educ

    Variable |        Obs        Mean    Std. dev.       Min        Max
-------------+---------------------------------------------------------
        wage |        526    5.896103    3.693086        .53      24.98
       lwage |        526    1.623268    .5315382  -.6348783   3.218076
        educ |        526    12.56274    2.769022          0         18

. regress wage educ

      Source |       SS           df       MS      Number of obs   =       526
-------------+----------------------------------   F(1, 524)       =    103.36
```

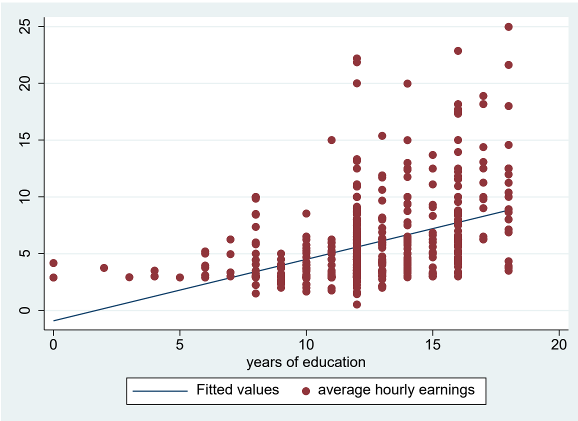
```
       Model |  1179.73204         1  1179.73204   Prob > F        =    0.0000
    Residual |  5980.68225       524  11.4135158   R-squared       =    0.1648
-------------+----------------------------------   Adj R-squared   =    0.1632
       Total |  7160.41429       525  13.6388844   Root MSE        =    3.3784

------------------------------------------------------------------------------
        wage | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
        educ |   .5413593    .053248    10.17   0.000     .4367534    .6459651
       _cons |  -.9048516   .6849678    -1.32   0.187    -2.250472    .4407687
------------------------------------------------------------------------------

.
. // prediction and confidence interval graph
. twoway lfitci wage educ, name(ci)
```



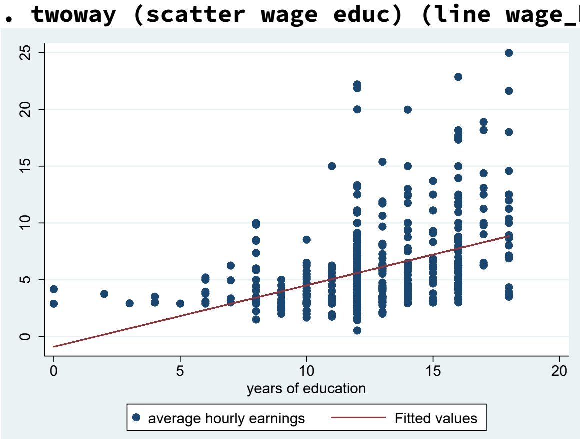

```
.
. // prediction and confidence interval graph + scatterplot of data
. twoway (lfitci wage educ) (scatter wage educ), name(scatter)
```



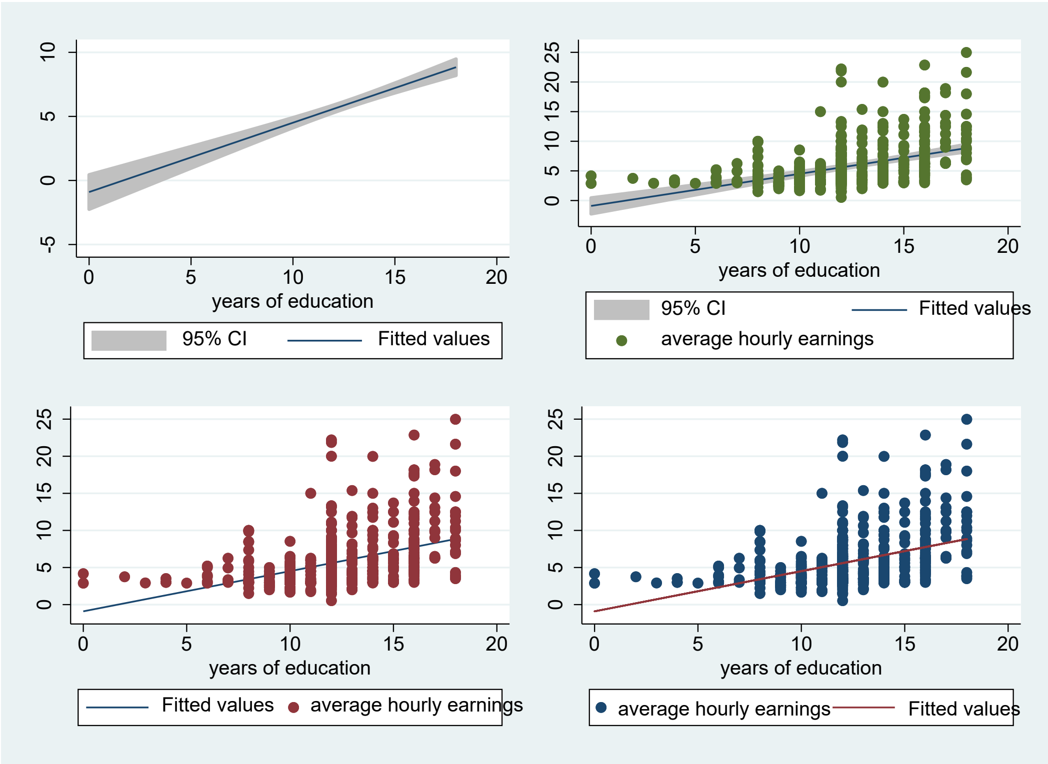

```
.
. // prediction w/o confidence interval graph + scatterplot of data
. twoway (lfit wage educ) (scatter wage educ), name(woci)
```




```
.
. // If I do the prediction graph myself
. regress wage educ

      Source |       SS           df       MS      Number of obs   =       526
-------------+----------------------------------   F(1, 524)       =    103.36
       Model |  1179.73204         1  1179.73204   Prob > F        =    0.0000
    Residual |  5980.68225       524  11.4135158   R-squared       =    0.1648
-------------+----------------------------------   Adj R-squared   =    0.1632
       Total |  7160.41429       525  13.6388844   Root MSE        =    3.3784

------------------------------------------------------------------------------
        wage | Coefficient  Std. err.      t    P>|t|     [95% conf. interval]
-------------+----------------------------------------------------------------
        educ |   .5413593    .053248    10.17   0.000     .4367534    .6459651
       _cons |  -.9048516   .6849678    -1.32   0.187    -2.250472    .4407687
------------------------------------------------------------------------------

. predict wage_hat
(option xb assumed; fitted values)

. twoway (scatter wage educ) (line wage_hat educ), name(oldfashion)
```




```
.
. // All four graphs in one for comparison
. graph combine ci scatter woci oldfashion
```



years of education
95% CI
Fitted values
average hourly earnings